

Gábor Hosu

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Mathematics and computer science background with hands-on experience in NLP, GPU programming, and ML research. I build AI systems from first principles, bridging research and production through clean, reproducible implementations.

Honours and Publications

Á. Kránics, G. Hosu , et al., "Multimodal RAG for Context-Aware Chatbots", IEEE SISY 23rd [DOI], [Preprint]	Oct 2025
2nd Prize at 28th Transylvanian Students' Scientific Conference	May 2025

Skills

Languages: Python, SQL (MS SQL)

ML/AI: PyTorch, Transformers, LangChain, Sentence-Transformers, Hugging Face

Data: NumPy, Pandas, Milvus **Backend/Infra:** FastAPI, SQLAlchemy, Docker, Azure, GitLab CI/CD

Other: Git, Linux **Familiar with:** OpenCL, Java, Spring, React, C++

Experience

Software Engineering Internship Codespring - Designed backend for a RAG chatbot: document ingestion, embedding, and retrieval pipeline. - Built and deployed REST APIs (auth, file management) via Docker and GitLab CI/CD to Azure.	Jul 2024 - Sept 2024
Fullstack mentorprogram Codespring Pre-internship training program covering fullstack development with Java, Spring, and React.	Oct 2023 - May 2024

Projects

Legal Document Form-Filling Prediction - Designed a lightweight ML model for legal document field-type prediction using text and layout context, targeting low-data settings where OCR-based approaches fail. - Built and annotated a dataset of 198 legal PDFs (19 classes); achieved 75% macro F1 (5-fold CV) with a hierarchical coarse-to-fine classification approach.	Apr 2026 - May 2026
Height Map Estimation From Normal Maps - Built GPU-based Poisson surface reconstruction pipeline from normal maps; derived and implemented finite-difference and FFT solvers from scratch. - Benchmarked iterative (Jacobi, Gauss-Seidel) and spectral methods in OpenCL + Python with Docker; evaluated scalability and accuracy on large-scale image grids.	Mar 2026 - May 2026
Embedding-Based Retrieval Paper Reproduction [Portfolio] [GitHub] [Paper] [HF (Models & Data)] - Reproduced a 2025 Google DeepMind paper on geometric limits of single-vector embedding retrieval under GPU constraints. - Built and evaluated synthetic IR pipelines (2K–50K docs), benchmarked quantized LLMs, tuned hyperparameters; presented findings at a research group seminar.	Nov 2025 - Jan 2026

Education

Babeş-Bolyai University , MSc, Data Analysis and Modelling Relevant coursework: NLP, Information Retrieval, Machine Learning, GPU Programming, Swarm Intelligence, Robotics	Oct 2025 - Present
Babeş-Bolyai University , BSc, Mathematics and Computer Science GPA: 9.58/10 Thesis: "A file-centric chatbot based on the RAG method"	2022 - 2025
Languages: English (Upper-Intermediate) Romanian (Intermediate) Hungarian (Native)	